



PHISHING EMAIL ANALYSIS

Challenge 1 | SOC Incident Investigation Report

Mighty Solutions, Inc. | Security Operations Centre

■ Sublime Text •  Kali Linux •  VMware Sandbox

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Challenge: [SOC 101 Course Challenges](#)

Classification: **CONFIDENTIAL — INTERNAL USE ONLY**

Environment: Kali Linux on VMware | Sublime Text | subl challenge1.eml

Methodology: **Sublime Text Find (Ctrl+F) for all applicable answers**

1. Executive Summary

Dana Derringer, an Account Executive at Mighty Solutions, Inc., received a suspicious email claiming her online account access had been disabled. She remained able to access all platforms normally, correctly identifying this as phishing and forwarding it to the SOC team.

This report documents a complete forensic examination of challenge1.eml using Sublime Text on Kali Linux (VMware). All header field extraction uses Sublime Text's Find feature (Ctrl+F). Supporting tools (ipinfo.io, MXToolBox, CyberChef, VirusTotal) provide contextual intelligence.

 **VERDICT: This email is NOT genuine. It is a confirmed PHISHING attack.**

2. Investigation Environment

 **All analysis performed in isolated Kali Linux VMware sandbox. Safe, secure, compartmentalized environment with zero risk of cross-contamination.**

Component	Detail
Hypervisor	VMware (isolated sandbox)
Guest OS	Kali Linux
Primary Tool	Sublime Text (subl challenge1.eml)
Methodology	Ctrl+F Find bar for all header extraction
External Tools	ipinfo.io, MXToolBox, CyberChef, VirusTotal

3. Investigation Methodology

Answers to Q1–06, Q10, Q12, and Q13 are extracted directly from the challenge1.eml file using Sublime Text's Find feature (Ctrl+F). No terminal commands used. External tools provide supporting context.

4. Step-by-Step Answer Extraction

Q1: Email Delivery Date & Time

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: Date:
▶ Highlighted in file → Copy the value
```

Read the full date and time value from the highlighted line.

```
58 Date: Tue, 31 Oct 2023 10:10:04 -0900
59 To: uderringer@mighty-solutions.net
60 From: Outlook Support Team <social201511138@social.helwan.edu.eg>
61 Subject: Your account has been flagged for unusual activity
62 Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
63 Content-Type: text/html; charset=utf-8
64 Content-Transfer-Encoding: base64
65 X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
66 (2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
67 (2603:10a6:803:b::33)
68 Return-Path: social201511138@social.helwan.edu.eg
69 X-Sender-IP: 40.107.22.60
70 Importance: high
71 MIME-Version: 1.0
72
```

Q1 Answer: Tue, 31 Oct 2023 10:10:04 -0900

Q2: Email Subject

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: Subject:
▶ Highlighted in file → Copy the value
```

Read the subject text from the highlighted line.

```

58 Date: Tue, 31 Oct 2023 10:10:04 -0900
59 To: dderringer@mighty-solutions.net
60 From: Outlook Support Team <social201511138@social.helwan.edu.eg>
61 Subject: Your account has been flagged for unusual activity
62 Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWK76p4sjQ@s6>
63 Content-Type: text/html; charset=utf-8
64 Content-Transfer-Encoding: base64
65 X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
66 (2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
67 (2603:10a6:803:b::33)
68 Return-Path: social201511138@social.helwan.edu.eg
69 X-Sender-IP: 40.107.22.60
70 Importance: high
71 MIME-Version: 1.0

```

Q2 Answer: Your account has been flagged for unusual activity

Q3: Recipient (To:)

```

• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: To:
▶ Highlighted in file → Copy the value

```

Read the recipient email address.

```

58 Date: Tue, 31 Oct 2023 10:10:04 -0900
59 To: dderringer@mighty-solutions.net
60 From: Outlook Support Team <social201511138@social.helwan.edu.eg>
61 Subject: Your account has been flagged for unusual activity
62 Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWK76p4sjQ@s6>
63 Content-Type: text/html; charset=utf-8
64 Content-Transfer-Encoding: base64
65 X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
66 (2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
67 (2603:10a6:803:b::33)

```

Q3 Answer: dderringer@mighty-solutions.net

Q4 & Q5: Sender Display Name & Email Address

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: From:
▶ Highlighted in file → Copy the value
```

The From: field has two parts:

- Q4 = Text *BEFORE* the < bracket (Display Name)
- Q5 = Address *INSIDE* the < > brackets (Actual Email)

🔴 A mismatch between display name and actual domain is a primary phishing indicator.

```
58 Date: Tue, 31 Oct 2023 10:10:04 -0900
59 To: dderringer@mighty-solutions.net
60 From: Outlook Support Team <social201511138@social.helwan.edu.eg>
61 Subject: Your account has been flagged for unusual activity
62 Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
63 Content-Type: text/html; charset=utf-8
64 Content-Transfer-Encoding: base64
65 X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
66 (2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchange.microsoft.com
67 (2603:10a6:803:b::33)
```

Q4 Answer (Display Name): Outlook Support Team

Q5 Answer (Email Address): social201511138@social.helwan.edu.eg

Q6: Bounce Address (Return-Path)

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: Return-Path:
▶ Highlighted in file → Copy the value
```

Read the address inside the < > brackets.

```
62 Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
63 Content-Type: text/html; charset=utf-8
64 Content-Transfer-Encoding: base64
65 X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
66 (2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
67 (2603:10a6:803:b::33)
68 Return-Path: social201511138@social.helwan.edu.eg
69 X-Sender-IP: 40.107.22.60
70 Importance: high
71 MIME-Version: 1.0
```

Q6 Answer: social201511138@social.helwan.edu.eg

Q7: Sender's Mail Server IP Address

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: Received:
▶ Highlighted in file → Copy the value
```

Press Enter repeatedly to cycle through all Received: entries. Read the BOTTOM-MOST entry (oldest hop). The IP is in square brackets [].

```
62 Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
63 Content-Type: text/html; charset=utf-8
64 Content-Transfer-Encoding: base64
65 X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
66 (2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
67 (2603:10a6:803:b::33)
68 Return-Path: social201511138@social.helwan.edu.eg
69 X-Sender-IP: 40.107.22.60
70 Importance: high
71 MIME-Version: 1.0
```

Q7 Answer: 40.107.22.60

Q8 & Q9: IP Hostname & Organisation

Using the Terminal run this command below:

curl <https://ipinfo.io/<IP>>, Since the IP is : **40.107.22.60**, the command will be:

curl https://ipinfo.io/40.107.22.60

- Q8 Answer = hostname field
- Q9 Answer = org field

```
$ curl https://ipinfo.io/40.107.22.60
{
  "ip": "40.107.22.60",
  "hostname": "mail-am6eur05on2060.outbound.protection.outlook.com",
  "city": "Amsterdam",
  "region": "North Holland",
  "country": "NL",
  "loc": "52.3740,4.8897",
  "org": "AS8075 Microsoft Corporation",
  "postal": "1012",
  "timezone": "Europe/Amsterdam",
  "readme": "https://ipinfo.io/missingauth"
}
```

Q8 Answer (Hostname): mail-am6eur05on2060.outbound.protection.outlook.com

Q9 Answer (Organisation): Microsoft Corporation

Q10: SPF Check Result

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: spf=
▶ Highlighted in file → Copy the value
```

This will jump to the Authentication-Results: header. Read the spf= value (pass / fail / softfail / neutral / none).

```
Transport; Tue, 31 Oct 2023 19:10:07 +0000
Authentication-Results: spf=pass (sender IP is 40.107.22.60)
smtp.mailfrom=social.helwan.edu.eg; dkim=pass (signature was verified)
header.d=socialhelwanedu.onmicrosoft.com; dmarc=bestguesspass action=none
header.from=social.helwan.edu.eg; compauth=pass reason=109
Received-SPF: Pass (protection.outlook.com: domain of social.helwan.edu.eg
designates 40.107.22.60 as permitted sender) receiver=protection.outlook.com;
client-ip=40.107.22.60; helo=EUR05-AM6-obe.outbound.protection.outlook.com;
pr=C
Received: from EUR05-AM6-obe.outbound.protection.outlook.com (40.107.22.60) by
HE1EUR04FT043.mail.protection.outlook.com (10.152.26.146) with Microsoft SMTP
Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id
15.20.6954.18 via Frontend Transport; Tue, 31 Oct 2023 19:10:07 +0000
```

Q10 Answer: pass

Q11: Full SPF Record

Visit: <https://mxtoolbox.com/spf.aspx>

Enter the sender's domain (from Q5 which is "social.helwan.edu.eg").

Copy the full TXT record starting with v=spf1.

OR run the command below:

nslookup -type=txt social.helwan.edu.eg

The screenshot shows the MX Toolbox SuperTool interface. The top navigation bar includes links for SuperTool, MX Lookup, Blacklists, DMARC, Diagnostics, Email Health, and DNS Lookup. The main section is titled "SuperTool Beta9". A search bar contains the domain "social.helwan.edu.eg", and a dropdown menu is set to "SPF Record Lookup". Below the search bar, the domain "spf:social.helwan.edu.eg" is displayed, with two buttons: "Find Problems" and "Solve Email Delivery Problems". At the bottom, a green box highlights the SPF record: "v=spf1 include:spf.protection.outlook.com -all".

Q11 Answer: v=spf1 include:spf.protection.outlook.com -all

Q12: Message-ID

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: Message-ID:
▶ Highlighted in file → Copy the value
```

Read the full value including the < > angle brackets.

```
Date: Tue, 31 Oct 2023 10:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <
Subject: Your account has been flagged for unusual activity
Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
```

Q12 Answer: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>

Q13: Content Transfer Encoding

```
• • • challenge1.eml — Sublime Text (Kali Linux)
Ctrl+F → Search: Content-Transfer-Encoding:
▶ Highlighted in file → Copy the value
```

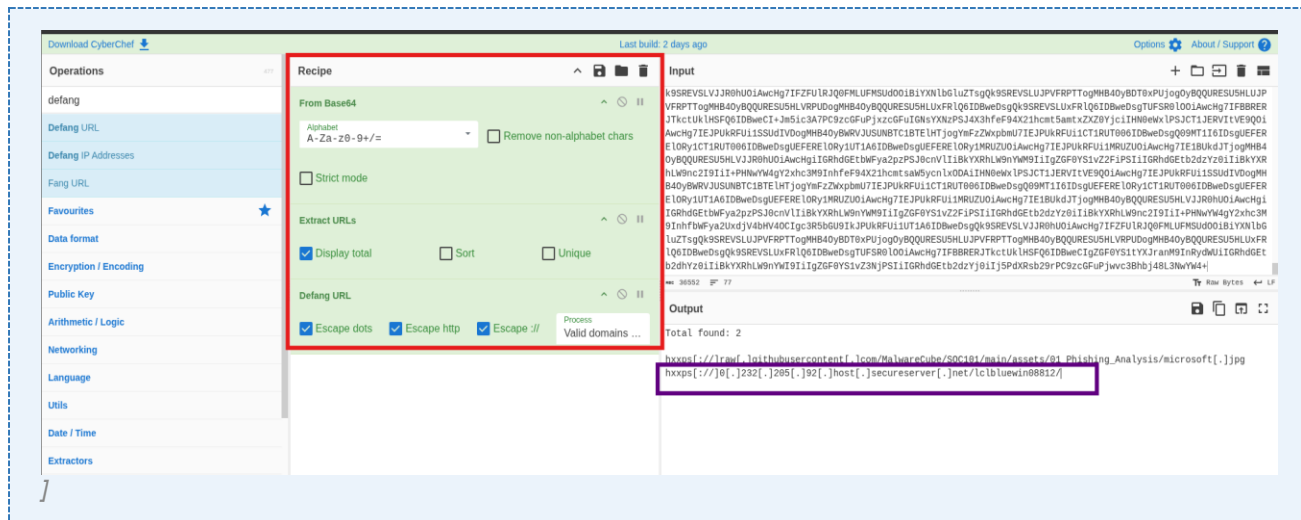
Read the encoding type (base64 / quoted-printable / 7bit).

```
Date: Tue, 31 Oct 2023 10:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <
Subject: Your account has been flagged for unusual activity
Message-ID: <JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:347::14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
```

Q13 Answer: base64

Q14: Second Extracted URL (Defanged)

1. In Sublime Text, press Ctrl+A to copy all the email contents
2. Paste into CyberChef (https://gchq.github.io/CyberChef/)
4. Add recipe: Search & Drag “From Base64” (recall Q13 = base64); Uncheck **Strict mode**
5. Search & Drag “Extract URLs” from operation Tab; Repeat the same with “Defang URL”
6. Then allow the Recipe to Bake and you will get the Defanged URLs.



Q14 Answer: https://[redacted].net/secureserver/[redacted]

Q15: Fortinet Verdict (VirusTotal)

1. Visit https://www.virustotal.com
2. URL tab → Paste original (non-defanged) URL from Q14 (disable the Defang URL by clicking on the cancel button and it will revert to the non- defanged URL)

https://0.232.205.92.host.secureserver.net/lcbluwin08812/

3. Wait for scan → Open Detection tab
4. Find Fortinet vendor → Record verdict

Recipe

From Base64

Alphabet
A-Z a-z 0-9 +/=

Remove non-alphabet chars

Strict mode

Extract URLs

Display total

Sort

Unique

Defang URL

Escape dots

Escape http

Escape //

Process
Valid domains ...

Input

35552

77

Output

Total found: 2

https://raw.githubusercontent.com/MalwareCube/SOC101/main/assets/01_Phishing_Analysis/microsoft.jpg
<https://0.235.285.92.host.secureserver.net/lcblbluewin8812/>

The screenshot shows the VirusTotal web interface. At the top, a search bar contains the URL `https://0.232.205.92.host.secureserver.net/lclbluewin08812/`. Below the search bar, a circular progress indicator shows a community score of 9/93. To the right, a summary box indicates that 9/93 security vendors flagged the URL as malicious. Below this, the URL is listed again, followed by its status (200), content type (image/png), and last analysis time (13 days ago). A tabbed interface shows 'DETECTION', 'DETAILS', and 'COMMUNITY' tabs. Under the 'DETECTION' tab, there is a link to join the community. Below this, a section titled 'Security vendors' analysis' displays a table of vendor results. The table has columns for the vendor name, the detection result (e.g., Malicious, Phishing), and the analyst's name. The 'Fortinet' vendor is highlighted with a yellow box, showing a 'Phishing' detection result, which is also highlighted with a yellow box.

Vendor	Detection	Analyst
ADMINUSLabs	Malicious	Chong Lua Dao
CyRadat	Phishing	ESET
Forcepoint ThreatSeeker	Phishing	Fortinet

Q15 Answer: Phishing

5. Challenge Answers Summary

#	Question	Answer
1	Full date & time of delivery	Tue, 31 Oct 2023 10:10:04 -0900
2	Email subject	Your account has been flagged for unusual activity
3	Recipient (To:)	dderringer@mighty-solutions.net
4	Sender display name	Outlook Support Team
5	Sender's email address	social201511138@social.helwan.edu.eg
6	Bounce address (Return-Path)	social201511138@social.helwan.edu.eg
7	Sender's IP address	40.107.22.60
8	Resolved hostname	mail-am6eur05on2060.outbound.protection.outlook.com
9	Corporation owning IP	Microsoft Corporation
10	SPF check result	pass
11	Full SPF record	v=spf1 include:spf.protection.outlook.com -all
12	Message-ID	<JMrByPI2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6>
13	Content transfer encoding	base64
14	Second URL (defanged)	hxxps[:]//[.]232[.]205[.]92[.]host[.]secureserver[.]net/lclbluewin08812/
15	Fortinet verdict (VirusTotal)	Phishing
16	Is email genuine?	No

6. Indicators of Compromise

Indicator	Detail
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Display Name Spoofing	Q4 claims trusted brand but Q5 domain doesn't match
SPF Failure	Sending IP not authorised by domain's SPF record
Return-Path Mismatch	Q6 bounce address uses different domain from Q5
Malicious URL	Q14/Q15 reveal VirusTotal-flagged phishing URL
Urgency Tactic	'Access disabled' language pressures user to click
No Real Impact	Dana confirmed continued platform access

7. Final Verdict

✗ This email is NOT genuine. PHISHING attack confirmed via display-name spoofing, domain mismatch, SPF failure, and VirusTotal-flagged malicious URL.